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Systematic Review of Ubiquilin (UBQLN1-4) Expression in Breast Cancer: Prognostic, Mechanistic, and Therapeutic Implications

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REVIEW TITLE AND BASIC DETAILS

Review title
Systematic Review of Ubiquilin (UBQLN1-4) Expression in Breast Cancer: Prognostic, Mechanistic, and Therapeutic Implications

Condition or domain being studied
Breast Cancer

Rationale for the review
Breast cancer remains a leading cause of cancer-related morbidity and mortality worldwide. Ubiquilins (UBQLN1-4) are proteins involved in proteostasis, apoptosis, and protein degradation pathways, and emerging evidence suggests they may influence tumor progression, survival outcomes, and therapeutic response. However, the prognostic significance and mechanistic role of ubiquilin in breast cancer remain unclear, and no systematic synthesis of existing clinical and preclinical evidence currently exists. This review aims to comprehensively evaluate the association between ubiquilin expression and clinical outcomes, elucidate underlying molecular mechanisms, and assess its potential as a predictive or prognostic biomarker and therapeutic target, thereby providing insights to guide future research and clinical management.

Review objectives
Review Objectives:

This systematic review aims to comprehensively evaluate the role of ubiquilin (UBQLN1–4) expression in breast cancer across clinical and preclinical settings. Specifically, the review will assess whether altered ubiquilin expression (high or low) is associated with differences in tumor aggressiveness, progression, and survival outcomes. In addition, it will synthesize mechanistic evidence on how ubiquilin dysregulation influences proteostasis, apoptosis, and metastasis, and explore its potential utility as a prognostic biomarker or therapeutic target.

Review Questions:

How does ubiquilin (UBQLN1–4) expression correlate with clinical outcomes in patients with breast cancer, including overall survival (OS), disease-free survival (DFS), and progression-free survival (PFS)?

What are the molecular and cellular mechanisms by which ubiquilin contributes to breast cancer initiation, progression, and metastasis?

Can ubiquilin serve as a prognostic or predictive biomarker, or a viable therapeutic target, in breast cancer management?

Keywords
Breast cancer; ubiquilin; Biomarkers; Prognosis; Survival; Proteostasis; Apoptosis; Metastasis; Therapeutic targets

Country
United States of America

ELIGIBILITY CRITERIA

Population

Included
Adults or pediatric patients diagnosed with breast cancer of any subtype, including estrogen receptor-positive (ER+), human epidermal growth factor receptor 2-positive (HER2+), and triple-negative breast cancer.

Patients at any stage of disease, including early-stage, locally advanced, or metastatic breast cancer.

Preclinical models relevant to breast cancer, including human breast cancer cell lines and animal models (e.g., murine xenografts).

Excluded
Studies focusing exclusively on non-breast cancers or non-cancerous conditions.

Case reports or case series with fewer than six participants.

Non-human models unrelated to breast cancer.

Intervention(s) or exposure(s)

Included

Laboratory procedure; Reverse transcription polymerase chain reaction; Others

This review includes studies that measure ubiquilin (UBQLN1–4) expression at the protein or mRNA level using validated molecular assays, such as immunohistochemistry or reverse transcription polymerase chain reaction (RT-PCR). It also includes studies that functionally manipulate ubiquilin through genetic methods (over expression, knockdown, knockout) or other laboratory-based interventions in preclinical models of breast cancer.

Excluded
Studies that do not directly measure or manipulate ubiquilin.

Studies evaluating ubiquilin in non-breast cancer contexts.

Clinical pharmacological interventions targeting unrelated pathways.

Reviews, editorials, commentaries, or case reports without original experimental data.

Comparator(s) or control(s)

Included
PICO tags selected: Usual Care

Comparators include low or absent ubiquilin expression in breast cancer samples, normal breast tissue controls, untreated groups in preclinical experiments, or standard-of-care treatment groups in clinical studies.

Study design

Both randomized and nonrandomized study types will be included.

Included
Randomised controlled trials (RCTs) assessing interventions that modulate ubiquilin expression in breast cancer.

Non-randomised studies, including cohort studies, case-control studies, and cross-sectional studies examining associations between ubiquilin expression and clinical outcomes.

Preclinical mechanistic studies, including experiments using human breast cancer cell lines or animal models (e.g., murine xenografts), that evaluate the functional role of ubiquilin in tumor progression, apoptosis, or metastasis.

Excluded
Case reports or case series with fewer than six participants.

Reviews, editorials, commentaries, or conference abstracts without original experimental data.

Studies investigating ubiquilin solely in non-cancerous contexts or in cancers other than breast cancer.

Context

This review will include studies conducted in clinical and preclinical settings relevant to breast cancer. Clinical studies may be conducted in hospitals, cancer centers, or population-based cohorts worldwide, including patients of any age and any breast cancer subtype or stage. Preclinical studies include experiments using human breast cancer cell lines or animal models (e.g., murine xenografts) that examine ubiquilin expression or functional modulation. No restrictions will be placed on geographic location, healthcare system, or socioeconomic context.

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO
05 November 2025

Review timeline
Start date: 14 October 2025. End date: 14 December 2025.

Date of registration in PROSPERO
10 November 2025

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol
A full protocol has been written and uploaded to PROSPERO. The protocol may be accessed through this link
https://www.crd.york.ac.uk/PROSPEROFILES/4225533b1bdaddf7d8a7f3b66c54eda4.pdf.

SEARCHING AND SCREENING

Search for unpublished studies
Only published studies will be sought.

Main sources that will be searched
The main databases to be searched are *CENTRAL - Cochrane Central Register of Controlled Trials*, *Embase.com*, *MEDLINE* and *PubMed*.

Other important or specialist databases that will be searched
The Cochrane Database of Systematic Reviews

Search language restrictions
The review will only include studies published in English.

Search date restrictions
There are no search date restrictions.

Other methods of identifying studies
No other methods will be used.

Link to search strategy
A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process
Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

Other relevant information about searching and screening
None

DATA COLLECTION PROCESS

Data extraction from published articles and reports
Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will be asked to provide any required data not available in published reports.

Study risk of bias or quality assessment
Risk of bias will be assessed using: *Cochrane RoB-2* and *ROBINS-I*

Risk of bias in non-randomized clinical studies will be assessed using ROBINS-I. Any randomized trials will be evaluated using Cochrane RoB-2. Preclinical studies will be assessed using a structured custom checklist covering experimental design, reproducibility, blinding, and outcome reporting.

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment
Risk of bias due to missing results will be assessed

Certainty assessment
The Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach will be used to assess the certainty of evidence for clinical outcomes from observational and randomized studies, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias. For preclinical studies, certainty will be assessed using a structured approach evaluating reproducibility, experimental rigor, sample size, blinding, and reporting quality, as no formal GRADE-equivalent exists for laboratory studies. Evidence will be rated as high, moderate, low, or very low certainty for each outcome.

OUTCOMES TO BE ANALYSED

Main outcomes

Overall survival (OS) in patients with breast cancer stratified by ubiquilin (UBQLN1–4) expression levels.

Disease-free survival (DFS) and progression-free survival (PFS) in relation to ubiquilin expression.

Tumor progression metrics, including size, grade, and metastasis rates.

Additional outcomes

Molecular pathway activation, including ubiquitin-proteasome signaling and apoptosis.

Response to chemotherapy or targeted therapies in relation to ubiquilin expression.

Prognostic or predictive potential of ubiquilin as a biomarker.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

Formal synthesis is planned for outcomes where data are sufficiently homogeneous (e.g., studies reporting survival outcomes with high vs low ubiquilin expression). Otherwise, findings will be summarized narratively.

Quantitative data from comparable studies (e.g., hazard ratios for overall survival or progression-free survival) will be pooled using random-effects meta-analysis.

Heterogeneity will be assessed using I^2 statistics and Cochran's Q test.

For studies that are not suitable for quantitative synthesis (e.g., preclinical mechanistic studies, differing assay methods, or small observational cohorts), results will be summarized narratively, highlighting trends and mechanistic insights.

CURRENT REVIEW STAGE

Stage of the review at this submission 1 change

Review stage	Started	Completed
Pilot work	✓	✓
Formal searching/study identification	✓	✓
Screening search results against inclusion criteria	✓	✓
Data extraction or receipt of IPD	✓	✓
Risk of bias/quality assessment	✓	✓
Data synthesis	✓	✓

Review status

The review is completed.

Publication of review results
Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members
Ms Katherine Brown. University of Louisville School of Medicine. United States of America.

No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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Funding source
Review has no funding and no agreed support from an academic institution and is done in authors' own time.

Peer review
There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Apoptosis; Biomarkers; Breast Neoplasms; Disease-Free Survival; Female; Humans; Immunohistochemistry; Neoplastic Processes; Prognosis; Progression-Free Survival; Proteostasis

Revision note 1 change

The review has been completed as of August 5, 2026.

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 14 October 2025). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Prognostic Value of ATM Expression Levels in Pancreatic Cancer: A Systematic Review and Meta-Analysis [published 14 July 2025] [CRD420251103815]. The review was judged **not to be similar**
- Talin-1: A Promising Prognostic Biomarker Across Multiple Cancer Types [published 27 May 2025] [CRD420251046105]. The review was judged **not to be similar**
- What is the prognostic and predictive value of NSUN2 expression across human solid tumors? A systematic review and meta-analysis. [published 15 June 2025] [CRD420251072935]. The review was judged **not to be similar**

PROSPERO version history 1 change

- Version 1.1, published 05 Aug 2026
- Version 1.0, published 10 Nov 2025

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